3.1 Argument and Validity

Arguments and argumentations. Loosely speaking, logic studies the principles ruling a correct *argument*. Just as we express propositions linguistically through declarative sentences, we express arguments linguistically through *argumentations* or *argument-texts*.

Notion 3.1 (Argument)**.** An **argument** is an arrangement of propositions, one of which, the **conclusion**, is indicated as following from the others, the **premises**.

Notion 3.2 (Argumentation)**.** An **argumentation** or **argument-text** is a linguistic expression that expresses an argument.

Since an argumentation $\mathtt{A}$ has to somehow express an arrangement of propositions, $\mathtt{A}$ may be construed as articulations of sentences, one of which expresses the conclusion of the argument expressed by $\mathtt{A}$, while the rest express its premises.

In natural language, we often use a *conclusion marker* to indicate that the sentence that follows expresses the conclusion of the argument expressed by $\mathtt{A}$. Consider, e.g., the following argumentation:

All humans are mortal. (Υ_1)
Socrates is human. (Υ_2)
Therefore, (Υ_3)
Socrates is mortal. (Υ_4)

Sentences (Υ_1) and (Υ_2) express the premises of the argument expressed by (Υ), while (Υ_4) expresses its conclusion. Sentence (Υ_4) is marked to express the conclusion via (Υ_3), a common *conclusion marker* in English.

In natural language, however, we often find argumentations with no explicit conclusion marker. For instance, it would suffice for sentences (Υ_1), (Υ_2), and (Υ_4) to be written or uttered in that exact order for the average English speaker to regard it as an argumentation whose conclusion is expressed by (Υ_4).

As with sentences and propositions, we will avoid some of the distinctions between arguments and argumentations. In particular, the following terminology will be convenient.

Notion 3.3 (Premises and conclusion of an argumentation)**.** Let $\mathtt{A}$ be an argumentation and let $\mathtt{P}_1, \ldots, \mathtt{P}_n$ stand for the premises and $\mathtt{C}$ for the conclusion of the argument $\mathtt{A}$ expresses. Hence, $\mathtt{P}_1, \ldots, \mathtt{P}_n$ are the **premises** of $\mathtt{A}$ and that $\mathtt{C}$ is its **conclusion**.

Historical note: Arguments

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>
> Author (XXXX–XXXX)

Standard form. In natural language, the premises and conclusions of an argumentation are not always easily identifiable. They are easy to identify in argumentation (♈), as it is written in a way approximating the standard form.

Notation 3.4 (Argumentation in standard form)**.** $\ulcorner \mathtt{P}_1 \ldots \mathtt{P}_n :\cdot\ \mathtt{C} \urcorner$ and $\ulcorner \mathtt{C}\ \cdot: \mathtt{P}_1 \ldots \mathtt{P}_n \urcorner$ both denote an argument, where $\mathtt{C}$ expresses the conclusion and $\mathtt{P}_1, \ldots, \mathtt{P}_n$ express the premises.

In notation 3.4, ':·' is a *right-conclusion marker*, because it marks the expression to its right as the one expressing conclusion. '·:' is instead a *left-conclusion marker*, as it marks the expression to its left as the one expressing the conclusion.*

*: Most authors use only a right-conclusion marker, which is normally '∴'. A few other authors also use '∵', serving as a left-conclusion marker.

Thus, argumentation (♈) may be expressed in standard form in any of the two following ways:

All humans are mortal. Socrates is human. :· Socrates is mortal.

Socrates is mortal. ·: All humans are mortal. Socrates is human.

It is also important to note that, in rewriting a given argumentation in standard form, we have some flexibility to rewrite the expressions contained in the original argumentation.

Consider, for example, the following argumentations:

All philosophers are human.	(♉$_1$)
All humans are mortal.	(♉$_2$)
Therefore,	(♉$_3$)
all philosophers are mortal.	(♉$_4$)

Humans die. (♊$_1$)
Philosophers are human. (♊$_2$)
Hence, (♊$_3$)
philosophers die. (♊$_4$)

Since they differ textually, argumentations (♉) and (♊) are different. The same holds for the pairs of expressions: (♉$_1$), (♊$_2$); (♉$_2$), (♊$_1$); (♉$_3$), (♊$_3$); and (♉$_4$), (♊$_4$). However, both argumentations express the same argument, and, for each of sentence pairs above, its two sentences express the same proposition.

Hence, it would be quite acceptable, when rewriting these argumentations in standard form, to do so in the same way.

All philosophers are human. All humans are mortal.
$:\cdot$ All philosophers are mortal. (♋)

Humans die. Philosophers are human.
$:\cdot$ Philosophers die. (♌)

In other words, it is quite acceptable to say that (♋) and (♌) are both rewritings of (♉) in standard form, as well as of (♌). After all, what we want to do with these standard forms is to more accurately express the argument expressed by the original argumentations, and not the original argumentations themselves.

Task 3.1. Write the following argumentations in standard form:

1. If it rains, the ground will get wet. And it's raining. Ergo, the ground is wet.
2. You should drink more water. For staying hydrated improves concentration and health.
3. Since the data are incomplete, the results are unreliable, as several variables were not considered.
4. I forgot my keys. Now I cannot enter the house.
5. Solar panels generate clean energy, so installing them is beneficial. After all, electricity bills are high.
6. The museum is worth visiting. It contains rare historical artefacts and interactive exhibits.
7. Chocolate contains sugar. However, eating sugar is unhealthy. Hence, eating chocolate must also be unhealthy.
8. Thus, the movie was long but entertaining.

Let us identify sentences the premises and conclusion of each argumentation above.

Argumentations 3.1.1 3.1.4, and, 3.1.7 are quite straightforward, as they are written almost in standard form Thus, 'ergo' and 'hence' work as right-conclusion markers in 3.1.1 and 3.1.7. Now, although we find no explicit conclusion marker in 3.1.4, we can easily identify the conclusion by the context.

Argumentations 3.1.2 and 3.1.6 are less typical, as they begin by stating the conclusion, and the premises follow as justification. In 3.1.2, 'for' works as some sort of premise marker. In 3.1.6, we have no such marker but the context suggests that the second sentence is the premise.

The forms of argumentations 3.1.3 and 3.1.5 are even less typical, the conclusion appearing in the middle. We have nevertheless some words indicating its premises and conclusions. Thus, in 3.1.3, 'since' seems to work as as a premise marker, while in 3.1.5, 'so' seems to work as a right-conclusion marker. However, it may be debatable whether these are proper argumentations at all.

Finally, 3.1.8 does not look like a proper argumentation, as it consists of a single sentence preceded by 'thus' – which often does work as a conclusion marker in English – with no premises. We will, however, regard 3.1.8 as an argumentation. For although argumentations without premises are rarely – if at all – expressed in colloquial language, they are crucial in logic.

Solution to task 3.1.

Following common practice, we use $\ulcorner \mathtt{P} \mathrel{{:}{\cdot}} \mathtt{C} \urcorner$ more often than $\ulcorner \mathtt{C} \mathrel{{\cdot}{:}} \mathtt{P} \urcorner$ to write an argumentation in standard form.

1. If it rains, the ground will get wet. It is raining.
   :· The ground is wet.
2. Staying hydrated improves concentration and health.
   :· You should drink more water.
3. The data are incomplete.
   Several variables were not considered.
   :· The results are unreliable.
4. I forgot my keys. :· I cannot enter the house.
5. Solar panels generate clean energy.
   Electricity bills are high.
   :· Installing solar panel is beneficial.
6. The museum contains rare historical artefacts and interactive exhibits. :· The museum is worth visiting.
7. Chocolate contains sugar. Eating sugar is unhealthy.
   :· Eating chocolate is unhealthy.
8. :· The movie was long but entertaining.

Historical note: The conclusion marker

Teutsche Algebra, title page

The usual conclusion marker '∴' was first introduced by Johann Heinrich Rahn (1622–1676) in his work *Teutsche Algebra oder algebraische Rechenkunst* (*German Algebra or the Art of algebraic Calculation*, 1659, pp. 52–3). The following is an extract:

> As $a\sqrt{bc}$ is to $d\sqrt{bc}$, so a is to d
>
> $$\therefore\ ad\sqrt{bc} = ad\sqrt{bc}.$$
>
> According to the rule of proportions, the product of the two outer terms is equal to the product of the two middle terms; this is here indicated by the three little dots ∴, which are called *ergo* or *therefore*.

Argumentations and validity. Note that argumentations, like sentences, are expressions conveying information which is, in some sense, complete – an argumentation expresses an argument that is either good or not.

This would suggest that argumentations fall within our notion of a sentence, and arguments within our notion of a proposition. This is, however, not the case. For arguments cannot be true or false.

Fact 3.5. Arguments cannot be true or false.

Fact 3.6. Arguments are not propositions.

Fact 3.7. Argumentations are not sentences.

What can be said to be true with respect to an argument is whether it is a good one. For instance, we can say that it is true that the argument expressed by an argumentation ⌜`P` :· `C`⌝ is good, e.g., in case the proposition expressed by `B` does follow from the proposition expressed by `A`.

To be more specific, there are at least two senses in which argument can be good.

Notion 3.8 (Argument validity)**.** A **valid argument** is one whose conclusion follows from its premises.

Notion 3.9 (Argument soundness)**.** A **sound argument** is one which is valid and whose premises are true.

We extend both notions to argumentations.

Notion 3.10 (Argumentation validity)**.** A **valid argumentation** is one which expresses a valid argument.

Notion 3.11 (Argumentation soundness)**.** A **sound argumentation** is one which expresses a sound argument.

Note, en passant, that the fact that something is an argument or an argumentation does not mean it is any good. It does not necessarily mean that there must be some logical connection between its premises and conclusion. It just means that it has a conclusion, and (zero or more) premises.

Logical validity. Not all arguments are valid for the same reasons. Thus, consider the following two arguments.

All swans are white. ∴ No swan is black. (♍)

All humans are mortal. All Greeks are humans.
∴ All Greeks are mortal. (♎)

Both arguments seem to be valid in the sense that its conclusion follows from their premises. Argument (♍), however, seems to be an instance of a schema which al of whose instances are valid arguments:

All `h` are `m`. All `g` are `h`. ∴ All `g` are `m` (♏)

Argument (♍), instead, does not seem to be come from an schema producing only valid arguments. It is valid because blackness and whiteness happen to be incompatible properties.

Thus, it seems that while argument (♍) is valid just because of meaning of the terms 'black' and 'white', argument (♎) by virtue of its logical form. In other words, argument (♎) is *logically valid*.

Definition 3.12 (Logical validity)**.** `A` is **logically valid** $\overset{\text{DF}}{=}$ every `B` having the same logical form is valid : `A`, `B` are both arguments or both argumentations.

The problem is now to decide when a given argument is valid or not. There is more than one view on the matter, and such views are often expressed via logical systems or systems of logic.

3.2 Logical Systems

A *system of logic* aka *logical system* is a mathematical structure intended to capture how expressions, or the information they express, entails or is entailed by others. In what follows, we will use a special notation to name logical system.

Notation 3.13 (Logical system)**.** A logical system is named with an expression starting with an upper-case letters in sans-serif font, such as: 'A', 'B', ..., 'Y', 'Z'.

Formally, we usually define a system of logic as consisting of a formal language – intended to express the logical form of statements – and a relation of logical consequence defined on that language – intended to express which inferences are logically valid or not.

Using the concepts introduced so far, we may express this definition as follows.

Definition 3.14 (Logical system)**.**

$$\langle \mathbf{Fo}_{\mathsf{S}}, \mathbf{Cn}_{\mathsf{S}} \rangle \text{ is a } \textbf{logical system} \overset{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S}} \subseteq \mathbf{Fo}$$
$$\mathbin{\&\!\!\&} \mathbf{Cn}_{\mathsf{S}} \subseteq \wp\mathbf{Fo}_{\mathsf{S}} \times \wp\mathbf{Fo}_{\mathsf{S}}$$

Here, $\langle \mathbf{Fo}_{\mathsf{S}}, \mathbf{Cn}_{\mathsf{S}} \rangle$ is a pair consisting of the formal language $\mathbf{Fo}_{\mathsf{S}}$, construed as a subset of $\mathbf{Fo}$, and a relation between $\mathbf{Cn}_{\mathsf{S}}$ between subsets of $\mathbf{Fo}_{\mathsf{S}}$. This very pair is our logical system. Thus, if we want the formula 'Pa' to be a logical consequence of the formula '$\forall x\, Px$' according to this system, we need to define '$\langle \mathbf{Fo}_{\mathsf{S}}, \mathbf{Cn}_{\mathsf{S}} \rangle$' so that $\langle \{\forall x\, Px\}, \{Pa\} \rangle \in \mathbf{Cn}_{\mathsf{S}}$.

So far, we have roughly presented the usual definition of a logical system. In this textbook, however, it will be convenient to analyse this notion in more detail.

In particular, for each logical system here considered, we will consider: its vocabulary $\mathbf{Vo}_{\mathsf{S}}$; and its inference relation $\mathbf{In}_{\mathsf{S}}$ (not to be confused with the consequence relation). These are all collections which may be sets or tuples depending on the what is convenient for the system to be described.

The vocabulary $\mathbf{Vo}_{\mathsf{S}}$ is just the collection of symbols we use in the system. It will be common that the vocabulary be of the form:

$$\mathbf{Vo}_{\mathsf{S}} = \left\langle \mathbf{Log}_{\mathsf{S}}, \mathbf{Non}_{\mathsf{S}}, \mathbf{Aux}_{\mathsf{S}} \right\rangle$$

where $\mathbf{Log}_{\mathsf{S}}$ contains the logical symbols, $\mathbf{Non}_{\mathsf{S}}$ the non-logical symbols, and $\mathbf{Aux}_{\mathsf{S}}$ the auxiliary symbols of S. From these symbols in $\mathbf{Vo}_{\mathsf{S}}$, we build the set of formulae $\mathbf{Fo}_{\mathsf{S}}$, which is the formal language of the system.

The inference relation $\mathbf{In}_{\mathsf{S}}$ collects all the possible ways formulae can be arranged into inferences in S – whether those inferences are valid or not. Finally, the consequence relation $\mathbf{Cn}_{\mathsf{S}} \subseteq \mathbf{In}_{\mathsf{S}}$ picks out the inferences that are actually valid in S. So while $\mathbf{In}_{\mathsf{S}}$ includes all possible inferences, $\mathbf{Cn}_{\mathsf{S}}$ contains only those that count as correct according to the system.

Two further notions are closely tied to logical systems. A *derivation system*, or *calculus*, is a collection of rules that tell us how to derive some formulae from others. An *interpretation system*, or *semantics*, assigns references to the logical symbols, and in this way determines what the formulae refer to or mean, under a given interpretation.

Let us now look at these systems in more detail using familiar logical systems as examples: the standard propositional logic, called 'S0' and the standard first-order logic, called 'S1'.

3.3 Vocabulary

Any logical system features a collection of logical symbols and a collection of non-logical symbols. We might also add a collection of auxiliary symbols, like, e.g., parentheses. The function of these symbols would be to regiment the expressions of our languages.

Thus, the vocabulary of a logical system S may be construed as a triple consisting of a collection $\mathbf{Log}_{\mathsf{S}}$ of logical symbols, a collection $\mathbf{Non}_{\mathsf{S}}$ of non-logical symbols, and a collection $\mathbf{Aux}_{\mathsf{S}}$ of auxiliary symbols.

Many of the systems will use some of the following alphabets:

Definition 3.15 (Basic alphabets)**.**

$$\begin{aligned} \mathbf{Abc} &\overset{\text{DF}}{=} \{`a\text{'}, `b\text{'}, `c\text{'}\}, & \mathbf{ABC} &\overset{\text{DF}}{=} \{`A\text{'}, `B\text{'}, `C\text{'}\}, \\ \mathbf{Pqr} &\overset{\text{DF}}{=} \{`p\text{'}, `q\text{'}, `r\text{'}\}, & \mathbf{PQR} &\overset{\text{DF}}{=} \{`P\text{'}, `Q\text{'}, `R\text{'}\}, \\ \mathbf{Xyz} &\overset{\text{DF}}{=} \{`x\text{'}, `y\text{'}, `z\text{'}\}, & \mathbf{XYZ} &\overset{\text{DF}}{=} \{`X\text{'}, `Y\text{'}, `X\text{'}\}, \\ \mathbf{Con} &\overset{\text{DF}}{=} \{`\neg\text{'}, `\vee\text{'}, `\wedge\text{'}, `\rightarrow\text{'}, `\leftrightarrow\text{'}\}, & \mathbf{Par} &\overset{\text{DF}}{=} \{`(\text{'}, `)\text{'}\}. \end{aligned}$$

These collections may assume different functions depending on the logical system in which they are used. However, we might explain their usual functions as follows:

- **Abc** serves for constructing individual constants or terms;
- **ABC** serves for constructing predicate constants;
- **Pqr** serves for constructing propositional variables;
- **PQR** serves for constructing predicate constants or propositional variables;
- **Xyz** serves for constructing individual variables;
- **XYZ** serves for constructing predicate variables;
- **Con** contains logical connectives;
- **Par** contains auxiliary signs for regimentation.

There is reason why the latter two sets *contain* their symbol types, while the others *serve for constructing* them. Normally – but not always – we just need a finite amount of logical symbols, while we normally need an infinite amount of non-logical symbols. Aristotle's canonical syllogistic is an exception to the latter.

Vocabulary of $\mathsf{S0}$. The logical symbols of the standard propositional logic are logical connectives, which usually are: '$\neg$' (negation), '$\vee$' (disjunction), '$\wedge$' (conjunction), '$\rightarrow$' (conditional), '$\leftrightarrow$' (biconditional), which are precisely the members of **Con**.

It would be nevertheless useful to construe the set $\mathbf{Log}_{\mathsf{S0}}$ of logical symbols of $\mathsf{S0}$, whose first member consists of the one-place connective '$\neg$', and the second of all the other connectives, which are two-place connectives.

Task 3.2. Define '$\mathbf{Log}_{\mathsf{S0}}$'.

Hint: Define '$\mathbf{Log}_{\mathsf{S0}}[1]$' and '$\mathbf{Log}_{\mathsf{S0}}[2]$' separately.

Definition 3.16 (Logical symbols of $\mathsf{S0}$ (answer to task 3.3))**.**

$$\mathbf{Log}_{\mathsf{S0}}[1] \overset{\text{DF}}{=} \{`\neg'\},$$
$$\mathbf{Log}_{\mathsf{S0}}[2] \overset{\text{DF}}{=} \mathbf{Con} - \{`\neg'\}.$$

The non-logical symbols of $\mathbf{Fo}_{\mathsf{S0}}$ are propositional variables, which are usually represented with the lower case letters 'p', 'q', 'r' and their subindexed variants 'p_1', 'p_2', ..., 'q_1', 'q_2', ... 'r_1', 'r_2', ...

Thus, we may define our set of non-logical symbols $\mathbf{Non}_{\mathsf{S0}}$ from **Pqr** and the set of numerals $\mathbf{Num}_{\mathbb{W}}$.

Task 3.3. Define '$\mathbf{Non}_{\mathsf{S0}}$'.

Definition 3.17 (Non-logical symbols of $\mathsf{S0}$ (answer to task 3.3))**.**

$$\mathbf{Non}_{\mathsf{S0}} \overset{\text{DF}}{=} \{\mathtt{p}_\mathtt{i} : \mathtt{p} \in \mathbf{Pqr} \mathbin{\wedge\!\!\!\wedge} \mathtt{i} \in \mathbf{Num}_{\mathbb{W}}\}$$

Thus, the vocabulary $\mathbf{Vo}_{\mathsf{S0}}$ may be defined as follows:

Definition 3.18 (Vocabulary of $\mathsf{S0}$)**.**

$$\mathbf{Aux}_{\mathsf{S0}} \overset{\text{DF}}{=} \mathbf{Par},$$
$$\mathbf{Vo}_{\mathsf{S0}} \overset{\text{DF}}{=} \left\langle \mathbf{Log}_{\mathsf{S0}}, \mathbf{Non}_{\mathsf{S0}}, \mathbf{Aux}_{\mathsf{S0}} \right\rangle.$$

Vocabulary of S1. The vocabulary to the standard first-order logic S1 is similar to that of S0. However, instead of propositional variables, we have individual terms and predicates. Moreover, we include two further logical symbols, the quantifiers '∃' and '∀'.

Given the complexity of this vocabulary, it may be convenient to define its logical and non-logical vocabularies as tuples.

The logical vocabulary $\mathbf{Vo}_{\mathsf{S1}}$ may be defined as a pair consisting of the connectives plus the quantifiers.

Task 3.4. Define '$\mathbf{Log}_{\mathsf{S1}}$'.

Hint: Define '$\mathbf{Log}_{\mathsf{S1}}[1]$' similarly as '$\mathbf{Log}_{\mathsf{S0}}$'.

Definition 3.19 (Non-logical symbols of S1 (answer to task 3.4)).

$$\begin{aligned}
\mathbf{Con}_{\mathsf{S1}}[1] &\overset{\text{DF}}{=} \{\text{`}\neg\text{'}\}, \\
\mathbf{Con}_{\mathsf{S1}}[2] &\overset{\text{DF}}{=} \mathbf{Con} - \{\text{`}\neg\text{'}\}, \\
\mathbf{Qua}_{\mathsf{S1}} &\overset{\text{DF}}{=} \{\text{`}\exists\text{'}, \text{`}\forall\text{'}\}, \\
\mathbf{Log}_{\mathsf{S1}} &\overset{\text{DF}}{=} \langle \mathbf{Con}_{\mathsf{S1}}, \mathbf{Qua}_{\mathsf{S1}} \rangle .
\end{aligned}$$

The second member of $\mathbf{Vo}_{\mathsf{S1}}$ will also be a pair, consisting of the individual terms $\mathbf{Ind}_{\mathsf{S1}}$ plus the predicates $\mathbf{Pre}_{\mathsf{S1}}$. However, as we have various types of individual terms and predicates, it may be convenient to codify these also as tuples.

Now, individual terms are either individual constants or individual variables, which are letters and subindexed letters. Thus, we might define a set of individual constants $\mathbf{ICon}_{\mathsf{S1}}$ and a set of individual variables $\mathbf{IVar}_{\mathsf{S1}}$ from the sets **Abc** and **Xyz**, respectively.

Task 3.5. Define '$\mathbf{ICon}_{\mathsf{S1}}$', '$\mathbf{IVar}_{\mathsf{S1}}$', and '$\mathbf{Ind}_{\mathsf{S1}}$'.

Definition 3.20 (Individual terms of S1 (answer to task 3.5)).

$$\begin{aligned}
\mathbf{ICon}_{\mathsf{S1}} &\overset{\text{DF}}{=} \{\mathtt{a}_{\mathtt{i}} : \mathtt{a} \in \mathbf{Abc} \wedge\!\!\wedge \mathtt{i} \in \mathbf{Num}_{\mathbb{W}}\}, \\
\mathbf{IVar}_{\mathsf{S1}} &\overset{\text{DF}}{=} \{\mathtt{x}_{\mathtt{i}} : \mathtt{x} \in \mathbf{Xyz} \wedge\!\!\wedge \mathtt{i} \in \mathbf{Num}_{\mathbb{W}}\}, \\
\mathbf{Ind}_{\mathsf{S1}} &\overset{\text{DF}}{=} \langle \mathbf{ICon}_{\mathsf{S1}}, \mathbf{IVar}_{\mathsf{S1}} \rangle .
\end{aligned}$$

The set of predicates $\mathbf{Pre}_{\mathsf{S1}}$ is a bit more complicated, as we need predicates of every arity – that is, one-place, two-place, etc. – and infinitely many of each of those.

Thus, one-place predicates will be the letters in $\mathbf{Pre}_{\mathsf{S1}}$ plus the super-index '^{1}' – i.e., by 'P^1', 'Q^1', 'R^1' – plus their subindexed versions – i.e., 'P^1_1', ..., 'Q^1_1', ..., 'R^1_1', etc.

In general, n-place predicates will be of the forms $\ulcorner P^{\mathtt{n}} \urcorner$, $\ulcorner Q^{\mathtt{n}} \urcorner$, $\ulcorner R^{\mathtt{n}} \urcorner$, where $\mathtt{n}$ is the numeral that stands for the number n, and of the forms of the subindexed variants of these expressions – i.e., $\ulcorner P_1^{\mathtt{m}} \urcorner$, ..., $\ulcorner Q_1^{\mathtt{m}} \urcorner$, ..., $\ulcorner R_1^{\mathtt{m}} \urcorner$, etc.

To organise this, we will define a set for each arity. Thus, $\mathbf{Pre}_1$ will contain our one-place predicates, $\mathbf{Pre}_2$ will contain our two-place predicates. In general, $\mathbf{Pre}_{\mathtt{n}}$ will contain our n-place predicates, where $\mathtt{n}$ is the numeral corresponding to n.

Task 3.6. Define $\ulcorner \mathbf{Pre}_{\mathtt{n}} \urcorner$ for each $\mathtt{n} \in \mathbf{Num}_{\mathbb{W}}$.

Hint: Write a schematic definition $\mathbf{Pre}_{\mathtt{n}}$ for $\mathtt{n} \in \mathbf{Num}_{\mathbb{W}}$.

Definition 3.21 (Predicates (answer to task 3.6)).

$$\mathbf{Pre}_{\mathtt{n}} \overset{\text{DF}}{=} \{\mathtt{P}_{\mathtt{i}}^{\mathtt{n}} : \mathtt{P} \in \mathbf{Pre} \curlywedge \mathtt{i} \in \mathbf{Num}_{\mathbb{W}}\} : \mathtt{n} \in \mathbf{Num}_{\mathbb{N}}.$$

Now we can define $\mathbf{Pre}_{\mathsf{S1}}$ as the tuple containing each of the sets defined above in the order expected.

Task 3.7. Define the '$\mathbf{Pre}_{\mathsf{S1}}$'.

Definition 3.22 (Predicates (answer to task 3.7)).

$$\mathbf{Pre}_{\mathsf{S1}}[\mathtt{n}] \overset{\text{DF}}{=} \mathbf{Pre}_{\mathtt{n}}$$
$$: \mathtt{n} \in \mathbf{Num}_{\mathbb{N}}.$$

Now we can define the tuples $\mathbf{Non}_{\mathsf{S1}}$ and $\mathbf{Vo}_{\mathsf{S1}}$ consisting of the non-logical symbols and full vocabulary of S1, respectively.

Task 3.8. Define (a) '$\mathbf{Non}_{\mathsf{S1}}$' and (b) '$\mathbf{Vo}_{\mathsf{S1}}$'.

Definition 3.23 (Non-logical symbols of S1 (answer to task 3.8.a)).

$$\mathbf{Non}_{\mathsf{S1}} \overset{\text{DF}}{=} \langle \mathbf{Ind}_{\mathsf{S1}}, \mathbf{Pre}_{\mathsf{S1}} \rangle .$$

Definition 3.24 (Vocabulary of S1 (answer to task 3.8.b)).

$$\mathbf{Aux}_{\mathsf{S1}} \overset{\text{DF}}{=} \mathbf{Par},$$
$$\mathbf{Vo}_{\mathsf{S1}} \overset{\text{DF}}{=} \left\langle \mathbf{Log}_{\mathsf{S1}}, \mathbf{Non}_{\mathsf{S1}}, \mathbf{Aux}_{\mathsf{S1}} \right\rangle .$$

And this is a way we may systematically construct the vocabulary of the standard logical systems.

3.4 Formulae

A formal language is articulated as a set of formulae. We need is a set of formulae of some specific forms, all of which are amenable to a certain logical treatment. The set of formulae $\mathbf{Fo}_{\mathsf{S}}$ of a logical system S is constructed from the symbols of its vocabulary $\mathbf{Vo}_{\mathsf{S}}$.

Formulae of $\mathsf{S0}$**.** The set of formulae $\mathbf{Fo}_{\mathsf{S0}}$ of the standard propositional logic are generated *recursively*. They are built up from simpler expressions according to precise formation rules.

$\mathbf{Fo}_{\mathsf{S0}}$ is based on the set $\mathbf{Non}_{\mathsf{S0}}$ (i.e., $\mathbf{Vo}_{\mathsf{S0}}[2]$) of *propositional variables*, which will be our atomic formulae. More complex formulae are then constructed with the help of the sets $\mathbf{Log}_{\mathsf{S0}}$ (i.e., $\mathbf{Vo}_{\mathsf{S0}}[1]$) of *connectives*, and $\mathbf{Aux}_{\mathsf{S0}}$ of auxiliary signs (i.e., $\mathbf{Vo}_{\mathsf{S0}}[3]$).

Table 3.1 shows the main forms of propositional formulae together with their natural-language analogues. To define this language, we may use a recursive definition like the following:

Preliminary 3.25 (Formulae of propositional logic)**.** The set $\mathbf{Fo}_{\mathsf{S0}}$ is recursively defined as follows:

1. $\mathsf{p} \in \mathbf{Non}_{\mathsf{S0}} \Rightarrow \mathsf{p} \in \mathbf{X}$.
2. $\mathsf{p} \in \mathbf{X} \Rightarrow \ulcorner\neg\mathsf{p}\urcorner \in \mathbf{X}$.
3. $\mathsf{p}, \mathsf{q} \in \mathbf{X} \Rightarrow \ulcorner(\mathsf{p} \vee \mathsf{q})\urcorner, \ulcorner(\mathsf{p} \wedge \mathsf{q})\urcorner, \ulcorner(\mathsf{p} \rightarrow \mathsf{q})\urcorner, \ulcorner(\mathsf{p} \leftrightarrow \mathsf{q})\urcorner \in \mathbf{X}$.
4. $\mathbf{Fo}_{\mathsf{S0}}$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define '$\mathbf{Fo}_{\mathsf{S0}}$' more shortly using '$\stackrel{\text{DF}}{=}$'.

Task 3.9. Define $\mathbf{Fo}_{\mathsf{S0}}$ using '$\stackrel{\text{DF}}{=}$'.

Definition 3.26 (Formulae of $\mathsf{S0}$ (answer to task 3.9))**.**

$$\begin{aligned}\mathbf{Fo}_{\mathsf{S0}} \stackrel{\text{DF}}{=} \mathbf{Non}_{\mathsf{S0}} \cup \\ \{\ulcorner\neg\mathsf{p}\urcorner : \mathsf{p} \in \mathbf{Fo}_{\mathsf{S0}}\} \cup \\ \{\ulcorner(\mathsf{p} \heartsuit \mathsf{q})\urcorner : \mathsf{p}, \mathsf{q} \in \mathbf{Fo}_{\mathsf{S0}} \mathbin{\wedge\!\!\!\wedge} \heartsuit \in \mathbf{Con}_{\mathsf{S0}}[2]\}.\end{aligned}$$

Table 3.1: Propositional formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Atomic	p	Peirce is a logician.
Atomic	q	Quine is a philosopher.
Negation	$\neg p$	It is not the case that Peirce is a teacher.
Disjunction	$(p \vee q)$	Peirce is a logician or Quine is a philosopher.
Conjunction	$(p \wedge q)$	Peirce is a logician and Quine is a philosopher.
Conditional	$(p \rightarrow q)$	If Peirce is a logician, then Quine is a philosopher.
Biconditional	$(p \leftrightarrow q)$	Peirce is a logician iff Quine is a philosopher.

Formulae of S1. $\mathbf{Fo}_{\mathsf{S1}}$ shares important features with $\mathbf{Fo}_{\mathsf{S1}}$. Their atomic formulae, however, are not propositional variables, but are more complexly built from predicates, which denote properties or relations, and individual terms, which denote specific or unspecified entities. We furthermore have *quantifiers* to specify how many unspecified entities are begin referred.

Table 3.2 shows some typical forms of first-order formulae together with their natural-language analogues.

A first useful step towards constructing $\mathbf{Fo}_{\mathsf{S1}}$ is to construct a set $\mathbf{ITer}_{\mathsf{S1}}$ of individual terms S1, containing the individual constants and individual variables of its language.

Definition 3.27 (Individual terms of S1).

$$\mathbf{ITer}_{\mathsf{S1}} \overset{\text{DF}}{=} \mathbf{ICon}_{\mathsf{S1}} \cup \mathbf{IVar}_{\mathsf{S1}}$$

In other words, $\mathbf{ITer}_{\mathsf{S1}}$ is the set of individual constants and variables of S1, or:

$$\mathbf{ITer}_{\mathsf{S1}} = \{`a', `a_0', \ldots, `c', `c_0', \ldots\} \cup \{`x', `x_0', \ldots, `z', `z_0', \ldots\}.$$

This makes it easy to define '$\mathbf{Fo}_{\mathsf{S1}}$' recursively.

Preliminary 3.28 (Formulae of $\mathbf{Fo}_{\mathsf{S1}}$). Where $\mathtt{n} \in \mathbf{Num}_{\mathbb{W}}$, the set $\mathbf{Fo}_{\mathsf{S1}}$ is recursively defined as follows:

1. $\mathtt{P} \in \mathbf{Pre}_{\mathtt{n}} \curlywedge \mathtt{t}_1, \ldots, \mathtt{t}_{\mathtt{n}} \in \mathbf{ITer}_{\mathsf{S1}} \Rightarrow \mathtt{P}\mathtt{t}_1 \ldots \mathtt{t}_{\mathtt{n}} \in \mathbf{X} : n \in \mathbb{N}$.
2. $\mathtt{F} \in \mathbf{X} \Rightarrow \ulcorner \neg \mathtt{F} \urcorner \in \mathbf{X}$.
3. $\mathtt{F}, \mathtt{G} \in \mathbf{X} \Rightarrow \ulcorner(\mathtt{F} \vee \mathtt{G})\urcorner, \ulcorner(\mathtt{F} \wedge \mathtt{G})\urcorner, \ulcorner(\mathtt{F} \rightarrow \mathtt{G})\urcorner, \ulcorner(\mathtt{F} \leftrightarrow \mathtt{G})\urcorner \in \mathbf{X}$.
4. $\mathtt{F} \in \mathbf{X} \curlywedge \mathtt{x} \in \mathbf{IVar}_{\mathsf{S1}} \Rightarrow \ulcorner \exists \mathtt{x}\, \mathtt{F} \urcorner, \ulcorner \forall \mathtt{x}\, \mathtt{F} \urcorner \in \mathbf{X}$.
5. $\mathbf{Fo}_{\mathsf{S1}}$ is the intersection of all sets $\mathbf{X}$ satisfying 1–4.

As with '$\mathbf{Fo}_{\mathsf{S0}}$', we might define '$\mathbf{Fo}_{\mathsf{S1}}$' more shortly using '$\overset{\text{DF}}{=}$'.

Task 3.10. Define '$\mathbf{Fo}_{\mathsf{S1}}$' using '$\overset{\text{DF}}{=}$'.

Table 3.2: First-order formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Atomic	Pa	Aristotle is a philosopher.
Atomic	Px	x is a philosopher.
Atomic	Rab	Aristotle respects Boole.
Atomic	Rxy	x respects y.
Existential	$\exists x\, Px$	Some object is a philosopher.
Universal	$\forall x\, Px$	Every object is a philosopher.

Definition 3.29 (Formulae of S1 (answer to task 3.10)).

$$\begin{aligned}\mathbf{Fo}_{\mathsf{S1}} \overset{\mathrm{DF}}{=} &\{\mathtt{Pt}_1 \ldots \mathtt{t}_n : \mathtt{P} \in \mathbf{Pre}_{\mathsf{S1}}[\mathtt{n}] \ggg \mathtt{t}_1 \ldots \mathtt{t}_n \in \mathbf{ITer}_{\mathsf{S1}}\} \cup \\ &\{\ulcorner \neg \mathtt{F} \urcorner : \mathtt{F} \in \mathbf{Fo}_{\mathsf{S1}}\} \cup \\ &\{\ulcorner (\mathtt{F} \heartsuit \mathtt{G}) \urcorner : \mathtt{F}, \mathtt{G} \in \mathbf{Fo}_{\mathsf{S1}} \ggg \heartsuit \in \mathbf{Con}_{\mathsf{S1}}[2]\} \cup \\ &\{\clubsuit \mathtt{vF} : \clubsuit \in \mathbf{Qua}_{\mathsf{S1}} \ggg \mathtt{v} \in \mathbf{IVar}_{\mathsf{S1}} \ggg \mathtt{F} \in \mathbf{Fo}_{\mathsf{S1}}\} \\ &: n \in \mathbb{N}\end{aligned}$$

3.5 Inference Relations

An inference relation $\mathbf{In}_{\mathsf{S}}$ is a set of inferences $\mathbf{I} = \langle \mathbf{A}, \mathbf{B} \rangle$. An inference is the formal counterpart of an argument, so $\mathbf{A}$ is a collection containing the formulae which represent the premises of the argument that $\mathbf{I}$ represents, while $\mathbf{B}$ contains the formulae representing its conclusions – normally only one.

Within a logical system S, $\mathbf{In}_{\mathsf{S}}$ represents all the forms in which formulae may be combined as premises and conclusions of an argument according to S.

In what follows, we will often use the inference bar notation, which will let us read our inferences with more easy.

Notation 3.30 (Inference bar)**.** The notations $\ulcorner \mathtt{A}_1, \ldots, \mathtt{A}_m \ / \ \mathtt{B}_1, \ldots, \mathtt{B}_n \urcorner$ and:

$$\frac{\begin{array}{c}\mathtt{A}_1 \\ \ldots \\ \mathtt{A}_m\end{array}}{\begin{array}{c}\mathtt{B}_1 \\ \ldots \\ \mathtt{B}_n\end{array}}$$

will denote an inference of one of the forms:

$$\begin{array}{ll}\langle \{\mathtt{A}_1, \ldots, \mathtt{A}_m\}, \{\mathtt{B}_1, \ldots, \mathtt{B}_n\} \rangle, & \langle \{\mathtt{A}_1, \ldots, \mathtt{A}_m\}, \langle \mathtt{B}_1, \ldots, \mathtt{B}_n \rangle \rangle, \\ \langle \langle \mathtt{A}_1, \ldots, \mathtt{A}_m \rangle, \{\mathtt{B}_1, \ldots, \mathtt{B}_n\} \rangle, & \langle \langle \mathtt{A}_1, \ldots, \mathtt{A}_m \rangle, \langle \mathtt{B}_1, \ldots, \mathtt{B}_n \rangle \rangle.\end{array}$$

Similarly, the notations $\ulcorner \mathtt{A}_1, \ldots, \mathtt{A}_m /\!\!/ \mathtt{B}_1, \ldots, \mathtt{B}_n \urcorner$ and:

$$\begin{array}{c}\mathtt{A}_1 \\ \ldots \\ \mathtt{A}_m \\ \hline\hline \mathtt{B}_1 \\ \ldots \\ \mathtt{B}_n\end{array}$$

will denote two inferences of the forms: $\ulcorner \mathtt{A}_1, \ldots, \mathtt{A}_m \ / \ \mathtt{B}_1, \ldots, \mathtt{B}_n \urcorner$ and $\ulcorner \mathtt{B}_1, \ldots, \mathtt{B}_n \ / \ \mathtt{A}_1, \ldots, \mathtt{A}_m \urcorner$.

This notation implicitly encloses each formula in corner marks.

Which of the four forms this notation represents is left to context and may be ambiguous.

It is important to remark here that it is formally possible for inferences to have multiple conclusions. This is because some logical systems allow for this. However, the inferences of most of the systems we will study will have only one formula as a conclusion, which we will represent as a singleton or a one-tuple.

Inference relation of modern logical systems. An inference in $\mathsf{S0}$, $\mathsf{S1}$, and virtually any modern logical system M is given between any (finite o infinite) collection of formulae representing the premises, and one formula, representing the conclusion.

In particular, in any given modern logical system M, $\langle \mathbf{A}, \{\mathtt{F}\}\rangle$ is an inference iff $\mathbf{A} \subseteq \mathbf{Fo}_{\mathsf{M}}$ and $\mathtt{F} \in \mathbf{Fo}_{\mathsf{M}}$.

Task 3.11. Define '$\mathbf{In}_{\mathsf{M}}$', where M is an unspecified modern logical system.

Definition 3.31 (Inference of M (answer to task 3.11))**.**

$$\mathbf{In}_{\mathsf{M}} \overset{\text{DF}}{=} \wp\mathbf{Fo}_{\mathsf{M}} \times \{\{\mathtt{F}\} : \mathtt{F} \in \mathbf{Fo}_{\mathsf{M}}\}$$
$$: \mathsf{M} \text{ is a standard logical system.}$$

Note that $\{\{\mathtt{F}\} : \mathtt{F} \in \mathbf{Fo}_{\mathsf{M}}\}$ is just the set of all singletons of $\mathbf{Fo}_{\mathsf{M}}$. Thus, if we wanted to simplify things, we could have just written:

$$\mathbf{In}_{\mathsf{M}} \overset{\text{DF}}{=} \wp\mathbf{Fo}_{\mathsf{M}} \times \mathbf{Fo}_{\mathsf{M}}.$$

However, since we want our definition of inference relation be so general as to include those logical systems which allow for multiple conclusions, we will have to use definition 3.31.

3.6 Consequence Relation

A consequence relation $\mathbf{Cn}_{\mathsf{S}}$ is always a subset of the set $\mathbf{In}_{\mathsf{S}}$ of inferences, which contains all and only the S-valid inferences – that is, those inferences which are logically valid in S.

Axiom 3.1 (Consequence relation)**.**

$$\mathbf{Cn}_{\mathsf{S}} \subseteq \mathbf{In}_{\mathsf{S}} : \mathsf{S} \text{ is a logical system.}$$

Note that the content of $\mathbf{Cn}_{\mathsf{S}}$, as we understand it here, needs not to depend on a specific semantics or calculus to be determined. We may, e.g., explicitly specify it by saying exactly which inferences are S-valid, as we will do later with A.

However, in most cases, including S0 and S1, it is necessary to specify that content by means of one such calculus or semantics.

Consequence relation of $\mathtt{S0}$. As we know, there are infinitely many $\mathtt{S0}$-valid inferences, which makes it impossible to list them. We have nevertheless a good idea of what we want $\mathbf{Cn}_{\mathtt{S0}}$ to be.

Among other things, we want all inferences of these forms to be $\mathtt{S0}$-valid. In other words, we want all the following inferences to be elements of $\mathbf{Cn}_{\mathtt{S0}}$.

Modus ponendo ponens (MPP): $(\mathtt{A} \rightarrow \mathtt{B}), \mathtt{A} \;/\; \mathtt{B}$.

Modus tollendo tollens (MTT): $(\mathtt{A} \rightarrow \mathtt{B}), \neg\mathtt{B} \;/\; \neg\mathtt{A}$.

Reiteration: $\mathtt{A} \;/\; \mathtt{A}$.

Disjunctive syllogism 1 (DS1): $(\mathtt{A} \vee \mathtt{B}), \neg\mathtt{A} \;/\; \mathtt{B}$.

Disjunctive syllogism 2 (DS2): $(\mathtt{A} \vee \mathtt{B}), \neg\mathtt{B} \;/\; \mathtt{A}$.

Law of excluded middle: $/\; (\mathtt{A} \vee \neg\mathtt{A})$.

Law of non-contradiction: $/\; \neg(\mathtt{A} \wedge \neg\mathtt{A})$.

Reflexivity of $\rightarrow$: $/\; (\mathtt{A} \rightarrow \mathtt{A})$.

Exportation: $((\mathtt{A} \wedge \mathtt{B}) \rightarrow \mathtt{C}) \;/\; (\mathtt{A} \rightarrow (\mathtt{B} \rightarrow \mathtt{C}))$.

Importation: $(\mathtt{A} \rightarrow (\mathtt{B} \rightarrow \mathtt{C})) \;/\; ((\mathtt{A} \wedge \mathtt{B}) \rightarrow \mathtt{C})$.

De Morgan 1: $\neg(\mathtt{A} \wedge \mathtt{B}) /\!\!/ (\neg\mathtt{A} \vee \neg\mathtt{B})$.

De Morgan 2: $\neg(\mathtt{A} \vee \mathtt{B}) /\!\!/ (\neg\mathtt{A} \wedge \neg\mathtt{B})$.

Double negation 1: $\mathtt{A} \;/\; \neg\neg\mathtt{A}$.

Double negation 2: $\neg\neg\mathtt{A} \;/\; \mathtt{A}$.

Idempotence of $\vee$: $\mathtt{A} /\!\!/ (\mathtt{A} \vee \mathtt{A})$.

Idempotence of $\wedge$: $\mathtt{A} /\!\!/ (\mathtt{A} \wedge \mathtt{A})$.

Simplification 1: $(\mathtt{A} \wedge \mathtt{B}) \;/\; \mathtt{A}$.

Simplification 2: $(\mathtt{A} \wedge \mathtt{B}) \;/\; \mathtt{B}$.

Addition 1: $\mathtt{A} \;/\; (\mathtt{A} \vee \mathtt{B})$.

Addition 2: $\mathtt{B} \;/\; (\mathtt{A} \vee \mathtt{B})$.

Conjunction: $\mathtt{A}, \mathtt{B} \;/\; (\mathtt{A} \wedge \mathtt{B})$.

Explosion (or ECQ): $(\mathtt{A} \wedge \neg\mathtt{A}) \;/\; \mathtt{B}$.

Commutativity of $\wedge$: $(\mathtt{A} \wedge \mathtt{B}) \;/\; (\mathtt{B} \wedge \mathtt{A})$.

Commutativity of $\vee$: $(\mathtt{B} \vee \mathtt{A}) \;/\; (\mathtt{A} \vee \mathtt{B})$.

Simply enumerating these or any other sets of rules instantiated, however, is not the most effective method for establishing $\mathbf{Cn}_{\mathtt{S0}}$. To do so, it may be better to define a semantics or a calculus from which we might define '$\mathbf{Cn}_{\mathtt{S0}}$'.

Historical note: Hilbert's positive logic

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> Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.
>
> Author (XXXX–XXXX)

Consequence relation of S1**.** Similar to S0, we cannot define the consequence relation of $\mathbf{Cn}_{\mathtt{S1}}$ by listing its elements one by one. We have infinitely many S1-valid inferences, which makes it impossible to enumerate them.

We nevertheless also have a good idea of what we want $\mathbf{Cn}_{\mathtt{S1}}$ to be. Among other things, where $\mathtt{F}, \mathtt{G} \in \mathbf{Fo}_{\mathtt{S1}}$ ⩕ $\mathtt{t} \in \mathbf{Ind}_{\mathtt{S1}}$ ⩕ $\mathtt{v}, \mathtt{w} \in \mathbf{IVar}_{\mathtt{S1}}$, we want all inferences of the forms below to be S1-valid, that is, to be elements of $\mathbf{Cn}_{\mathtt{S1}}$.

Universal instantiation:

$\forall \mathtt{v}\, \mathtt{F}[\mathtt{v}]\ /\ \mathtt{F}[\mathtt{t}]$: $\mathtt{t}$ is free for $\mathtt{v}$ in $\mathtt{F}$.

Existential generalisation:

$\mathtt{F}[\mathtt{t}]\ /\ \exists \mathtt{v}\, \mathtt{F}[\mathtt{v}]$: $\mathtt{t}$ is free for $\mathtt{v}$ in $\mathtt{F}$.

Quantifier equivalences:

- $\neg \forall \mathtt{v}\, \neg \mathtt{F}[\mathtt{v}] /\!\!/ \exists \mathtt{v}\, \mathtt{F}[\mathtt{v}]$.
- $\neg \exists \mathtt{v}\, \neg \mathtt{F}[\mathtt{v}] /\!\!/ \forall \mathtt{v}\, \mathtt{F}[\mathtt{v}]$.

Quantifier negation laws:

- $\neg \forall \mathtt{v}\, \mathtt{F}[\mathtt{v}] /\!\!/ \exists \mathtt{v}\, \neg \mathtt{F}[\mathtt{v}]$.
- $\neg \exists \mathtt{v}\, \mathtt{F}[\mathtt{v}] /\!\!/ \forall \mathtt{v}\, \neg \mathtt{F}[\mathtt{v}]$.

Universal distribution over conjunction:

$\forall \mathtt{v}\, (\mathtt{F}[\mathtt{v}] \wedge \mathtt{G}[\mathtt{v}]) /\!\!/ \forall \mathtt{v}\, \mathtt{F}[\mathtt{v}] \wedge \forall \mathtt{v}\, \mathtt{G}[\mathtt{v}]$.

Existential distribution over disjunction:

$\exists \mathtt{v}\, (\mathtt{F}[\mathtt{v}] \vee \mathtt{G}[\mathtt{v}]) /\!\!/ \exists \mathtt{v}\, \mathtt{F}[\mathtt{v}] \vee \exists \mathtt{v}\, \mathtt{G}[\mathtt{v}]$.

Universal quantifier commutation:

$\forall \mathtt{v}\, \forall \mathtt{w}\, \mathtt{F}[\mathtt{v}][\mathtt{w}] /\!\!/ \forall \mathtt{w}\, \forall \mathtt{v}\, \mathtt{F}[\mathtt{v}][\mathtt{w}]$.

Existential quantifier commutation:

$\exists \mathtt{v}\, \exists \mathtt{w}\, \mathtt{F}[\mathtt{v}][\mathtt{w}] /\!\!/ \exists \mathtt{w}\, \exists \mathtt{v}\, \mathtt{F}[\mathtt{v}][\mathtt{w}]$.

As with S0, simply enumerating these or any other sets of rules instantiated, however, is not the most effective method for establishing $\mathbf{Cn}_{\mathsf{S1}}$. To do so, it may be better to define a semantics or a calculus from which we might define '$\mathbf{Cn}_{\mathsf{S1}}$'.

3.7 Interpretation System

The formulae in a formal language $\mathbf{Fo}_{\mathsf{S}}$, normally, do not stand for propositions. In the formula 'P^1a', for instance, 'P^1' and 'a' stand for non-logical symbols without any established meaning. All we know of 'P^1' is that it stands for some one-place property, and of 'a', that it stands for some specific object in the domain.

For this reason, we can say that the formulae in $\mathbf{Fo}_{\mathsf{S}}$ are normally neither true nor false. However, given some further information about its non-logical symbols is supplied, a formula may be interpreted as being true or false.

Thus, if we interpret 'P^1' as standing for the predicate of being a philosopher and 'a' as standing for Aristotle, then 'P^1a' would stand for the proposition that Aristotle is a philosopher, making 'P^1' true under this interpretation.

In general, an interpretation $\mathfrak{I}$ on a formal language $\mathbf{Fo}_{\mathsf{S}}$ provides a mapping of non-logical symbols $\mathbf{Non}_{\mathsf{S}}$ of S into a domain.

Definition 3.32 (Interpretation)**.**

$$\begin{aligned}\mathfrak{I} \text{ is an interpretation on } \mathbf{Fo}_{\mathsf{S}} \overset{\text{DF}}{=} \mathfrak{I} = \langle \mathbf{D}_{\mathfrak{I}}, \iota_{\mathfrak{I}} \rangle \\ : \iota_{\mathfrak{I}} \subseteq \mathbf{Non}_{\mathsf{S}} \times \mathbf{D}_{\mathfrak{I}}.\end{aligned}$$

The non-logical symbols of a propositional logical system S are propositional variables which an interpretation will map into logical values, e.g. truth and falsity. In a first-order logic, those logical symbols are individual terms and predicates which are mapped into objects and sets in a domain.

In both cases, we then extend those interpretation into *valuations* from all formulae of $\mathbf{Fo}_{\mathsf{S}}$ into logical values.

Definition 3.33 (Valuation)**.**

$$\begin{aligned}\hat{\mathfrak{I}} \text{ is a valuation on } \mathbf{Fo}_{\mathsf{S}} \overset{\text{DF}}{=} \hat{\mathfrak{I}} = \langle \mathbf{LV}, \hat{\iota}_{\mathfrak{I}} \rangle \\ : \hat{\iota}_{\mathfrak{I}} \subseteq \mathbf{Fo}_{\mathsf{S}} \times \mathbf{LV} \ \curlywedge \\ \mathbf{LV} \text{ is a set of logical values.}\end{aligned}$$

An *interpretation system* or a *semantics* $\mathbf{Int}_{\mathsf{S}}$ on a logic system S consists of a set $\mathbf{Int}_{\mathsf{S}}$ of interpretations on $\mathbf{Int}_{\mathsf{S}}$, together with a set $\mathbf{LV}_{\mathsf{S}}$ of logical values, a subset $\mathbf{Des}^{+}_{\mathsf{S}}$ of $\mathbf{LV}_{\mathsf{S}}$ of designated logical values, a subset $\mathbf{Des}^{-}_{\mathsf{S}}$ of $\mathbf{LV}_{\mathsf{S}}$ of anti-designated logical values, a set $\mathbf{Val}_{\mathsf{S}}$ of valuations on $\mathbf{Fo}_{\mathsf{S}}$, and a semantic relation of logical consequence $\vDash_{\mathsf{S}}$.

Definition 3.34 (Interpretation system)**.**

$$\begin{aligned}&\mathbf{Sem}_{\mathsf{S}} \text{ is a semantics on } \mathsf{S} \\ &\quad \overset{\text{DF}}{=} \mathbf{Sem}_{\mathsf{S}} = \langle \mathbf{Int}_{\mathsf{S}}, \mathbf{LV}_{\mathsf{S}}, \mathbf{Val}_{\mathsf{S}}, \mathbf{Des}^{+}_{\mathsf{S}}, \mathbf{Des}^{-}_{\mathsf{S}}, \vDash_{\mathsf{S}} \rangle \\ &\quad : \mathbf{Int}_{\mathsf{S}} \text{ is a set of interpretations on } \mathbf{Fo}_{\mathsf{S}} \ \curlywedge \\ &\quad\ \ \mathbf{LV}_{\mathsf{S}} \text{ is a set of logical values } \curlywedge \\ &\quad\ \ \mathbf{Des}^{+}_{\mathsf{S}} \subseteq \mathbf{LV}_{\mathsf{S}} \ \curlywedge \\ &\quad\ \ \mathbf{Des}^{-}_{\mathsf{S}} \subseteq \mathbf{LV}_{\mathsf{S}} \ \curlywedge \\ &\quad\ \ \mathbf{Val}_{\mathsf{S}} \text{ is a set of valuations on } \mathbf{Fo}_{\mathsf{S}} \ \curlywedge \\ &\quad\ \ \vDash_{\mathsf{S}} \subseteq \mathbf{In}_{\mathsf{S}}.\end{aligned}$$

Intuitively, the set $\mathbf{Des}^{+}_{\mathsf{S}}$ of designated logical values contains those values related to the true or what can be asserted or accepted under an interpretation, while the set $\mathbf{Des}^{-}_{\mathsf{S}}$ of anti-designated values contains those values related to the false or what can be denied or rejected under an interpretation.

In order to present the semantics of propositional connectives in a more friendly way, we might use truth-tables.

Notation 3.35 (Truth tables)**.** The table

$\mathtt{A}$	$\bullet\mathtt{A}$
v_1	w_1
...	...
v_n	w_n

expresses that, under a given valuation $\hat{\mathfrak{I}}$, we have:

$$\begin{aligned} \iota_{\mathfrak{I}}(\mathtt{A}) = v_1 &\Rightarrow \hat{\iota}_{\mathfrak{I}}(\ulcorner \bullet \mathtt{A} \urcorner) = w_1 & \curlywedge \\ &\dots & \curlywedge \\ \iota_{\mathfrak{I}}(\mathtt{A}) = v_n &\Rightarrow \hat{\iota}_{\mathfrak{I}}(\ulcorner \bullet \mathtt{A} \urcorner) = w_n. \end{aligned}$$

Moreover, the tables:

$\mathtt{A}$	$\mathtt{B}$	$\mathtt{A} \heartsuit \mathtt{B}$
u_1	v_1	w_1
...	...	...
u_n	v_n	w_n

$\heartsuit$	v_1	...	v_n
u_1	w_1	...	w_a
...	...	...	...
u_n	w_b	...	w_n

both express that, under a given valuation $\hat{\mathfrak{I}}$, we have:

$$\begin{aligned} \iota_{\mathfrak{I}}(\mathtt{A}) = u_1 \curlywedge \iota_{\mathfrak{I}}(\mathtt{B}) = v_1 &\Rightarrow \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \heartsuit \mathtt{B})\urcorner) = w_1 & \curlywedge \\ &\dots & \curlywedge \\ \iota_{\mathfrak{I}}(\mathtt{A}) = u_n \curlywedge \iota_{\mathfrak{I}}(\mathtt{B}) = v_n &\Rightarrow \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \heartsuit \mathtt{B})\urcorner) = w_n \end{aligned}$$

Let us see how the interpretation systems of standard logical systems work.

Semantics for S0**.** In a standard propositional logical system S0, the non-logical symbols are propositional variables. A classical propositional interpretation, or cpi for short, assigns to each propositional variable a classical truth value, either **1** (truth) or **0** (falsity).

Definition 3.36 (Classical propositional interpretation)**.**

$$\langle \mathbf{D}_{\mathfrak{I}}, \iota_{\mathfrak{I}} \rangle \text{ is a } \mathbf{cpi} \text{ on } \mathbf{Fo}_{\mathrm{S}} \overset{\mathrm{DF}}{=} \mathbf{D}_{\mathfrak{I}} = \{\mathbf{0}, \mathbf{1}\} \curlywedge \iota_{\mathfrak{I}} : \mathbf{Pros}_{\mathrm{S}} \to \{\mathbf{0}, \mathbf{1}\}.$$

In other words, $\langle \mathbf{D}_{\mathfrak{I}}, \iota_{\mathfrak{I}} \rangle$ is a cpi on $\mathbf{Fo}_{\mathrm{S}}$ iff its domain is the set of logical values $\{\mathbf{0}, \mathbf{1}\}$ and its interpretation map $\iota_{\mathfrak{I}}$ is a function from the set of propositional variables to $\{\mathbf{0}, \mathbf{1}\}$ – that is, each propositional variable is assigned either **0** or **1**, but not both.

A cpi $\mathfrak{I}$ is extended to all formulae of $\mathbf{Fo}_{S0}$ by means of a valuation $\hat{\mathfrak{I}}$, which is its classical propositional extension (cpe).

Definition 3.37 (Classical propositional extension)**.** A valuation $\hat{\mathfrak{I}}$ is a **cpe** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S$

$$
\begin{aligned}
\overset{\text{DF}}{=} \ & \hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \iota_{\mathfrak{I}}(\mathtt{A}) && \wedge\!\!\wedge \\
& \hat{\iota}_{\mathfrak{I}}(\ulcorner\neg\mathtt{F}\urcorner) = \mathbf{1} \Leftrightarrow \hat{\iota}_{\mathfrak{I}}(\mathtt{F}) = \mathbf{0} && \wedge\!\!\wedge \\
& \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{F} \wedge \mathtt{G})\urcorner) = \mathbf{1} \Leftrightarrow \hat{\iota}_{\mathfrak{I}}(\mathtt{F}) = \mathbf{1} \wedge\!\!\wedge\ \hat{\iota}_{\mathfrak{I}}(\mathtt{G}) = \mathbf{1} && \wedge\!\!\wedge \\
& \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{F} \vee \mathtt{G})\urcorner) = \mathbf{1} \Leftrightarrow \hat{\iota}_{\mathfrak{I}}(\mathtt{F}) = \mathbf{1} \vee\!\!\vee\ \hat{\iota}_{\mathfrak{I}}(\mathtt{G}) = \mathbf{1} && \wedge\!\!\wedge \\
& \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{F} \rightarrow \mathtt{G})\urcorner) = \mathbf{1} \Leftrightarrow \hat{\iota}_{\mathfrak{I}}(\mathtt{F}) = \mathbf{0} \vee\!\!\vee\ \hat{\iota}_{\mathfrak{I}}(\mathtt{G}) = \mathbf{1} && \wedge\!\!\wedge \\
& \hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{F} \leftrightarrow \mathtt{G})\urcorner) = \mathbf{1} \Leftrightarrow \hat{\iota}_{\mathfrak{I}}(\mathtt{F}) = \hat{\iota}_{\mathfrak{I}}(\mathtt{G}) \\
& : \mathtt{A} \in \mathbf{Pro}_S \wedge\!\!\wedge\ \mathtt{F}, \mathtt{G} \in \mathbf{Fo}_S.
\end{aligned}
$$

Notation 3.35 makes it possible to rephrase definition 3.37 as follows:

Definition 3.38 (Classical propositional valuation)**.** A valuation $\hat{\mathfrak{I}}$ is a **classical extension** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S \overset{\text{DF}}{=} \hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \iota_{\mathfrak{I}}(\mathtt{A}) \wedge\!\!\wedge$

F	¬F
0	1
1	0

∧	0	1
0	0	0
1	0	1

∨	0	1
0	0	1
1	1	1

→	0	1
0	1	1
1	0	1

↔	0	1
0	1	0
1	0	1

$: \mathtt{A} \in \mathbf{Pro}_S \wedge\!\!\wedge\ \mathtt{F} \in \mathbf{Fo}_S.$

Now, as you might have guessed the one designated value in a semantics for S0 is $\mathbf{1}$, that is, truth; while the only anti-designated value is $\mathbf{0}$, that is, falsity. The standard notion of semantic logical consequence says that, $\mathtt{B}$ is a semantic logical consequence of $\mathtt{A}$ iff there is no interpretation $\mathfrak{I}$ such that $\hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \mathbf{1}$ but $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$.

We can generalise this standard notion of semantic logical consequence using the notions of designated and anti-designated value.

Definition 3.39 (Semantic consequence)**.** $\vDash_S$ is a standard semantic consequence relation $\overset{\text{DF}}{=}$

$$
\begin{aligned}
\langle \mathbf{A}, \{\mathtt{B}\}\rangle \in \vDash_S \Leftrightarrow \neg\exists\mathfrak{I}\,(&\mathfrak{I} \text{ is a cpi} \wedge\!\!\wedge \\
&\forall\!\!\forall \mathtt{A} \in \mathbf{A}\,(\hat{\iota}_{\mathfrak{I}}(\mathtt{A}) \in \mathbf{Des}_S^{+} \wedge\!\!\wedge\ \hat{\iota}_{\mathfrak{I}}(\mathtt{B}) \in \mathbf{Des}_S^{-})).
\end{aligned}
$$

We will adopt the following convention:

Notation 3.40.

$$\mathtt{A} \vDash_{\mathsf{S}} \mathtt{B} \overset{\text{DF}}{=} \ulcorner\langle \mathtt{A}, \mathtt{B}\rangle \in \vDash_{\mathsf{S}}\urcorner$$
$$\mathtt{A} \nvDash_{\mathsf{S}} \mathtt{B} \overset{\text{DF}}{=} \ulcorner\langle \mathtt{A}, \mathtt{B}\rangle \notin \vDash_{\mathsf{S}}\urcorner$$

Now we can define an interpretation system for S0, which we will call $\mathbf{Sem}_{\mathsf{S0}}$.

Definition 3.41 ($\mathbf{Sem}_{\mathsf{S0}}$)**.**

$$\begin{aligned}
&\mathbf{Sem}_{\mathsf{S0}} \text{ is a semantics on } \mathsf{S0} \\
&\overset{\text{DF}}{=} \mathbf{Int}_{\mathsf{S0}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a cpi on } \mathbf{Fo}_{\mathsf{S0}}\} && \curlywedge \\
&\quad \mathbf{LV}_{\mathsf{S0}} = \{\mathbf{0}, \mathbf{1}\} && \curlywedge \\
&\quad \mathbf{Des}^{+}_{\mathsf{S0}} = \{\mathbf{1}\} && \curlywedge \\
&\quad \mathbf{Des}^{-}_{\mathsf{S0}} = \{\mathbf{0}\} && \curlywedge \\
&\quad \mathbf{Val}_{\mathsf{S0}} = \{\hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{\mathsf{S0}}\, (\hat{\mathfrak{I}} \text{ is a cpe of } \mathfrak{I})\} && \curlywedge \\
&\quad \vDash_{\mathsf{S0}} \subseteq \mathbf{In}_{\mathsf{S0}} \curlywedge \vDash_{\mathsf{S0}} \text{ is classical.}
\end{aligned}$$

Task 3.12. Show that all inferences of the forms listed on p. 88 are members of $\vDash_{\mathsf{S0}}$.

As a previous remark, note $\vDash_{\mathsf{S0}}$ is classical. Hence, by definition 3.39, $\mathtt{A}_1, \ldots, \mathtt{A}_n \vDash_{\mathsf{S0}} \mathtt{B}$ iff there is no cpi $\mathfrak{I}$ such that $\hat{\iota}_{\mathfrak{I}}(\mathtt{A}_1) = \ldots = \hat{\iota}_{\mathfrak{I}}(\mathtt{A}_n) = \mathbf{1}$, but $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$.

We will show this for two of the cases required by task 3.12.

Theorem 3.42. $(\mathtt{A} \longrightarrow \mathtt{B}), \mathtt{A} \vDash_{\mathsf{S0}} \mathtt{B}$ – i.e., $(\mathtt{A} \longrightarrow \mathtt{B}), \mathtt{A} \,/\, \mathtt{B} \in \vDash_{\mathsf{S0}}$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{S0}}$ and assume, for reductio, that there is a cpi $\langle\{\mathbf{0}, \mathbf{1}\}, \iota_{\mathfrak{I}}\rangle$ such that $\hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \longrightarrow \mathtt{B})\urcorner) = \hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \mathbf{1}$, but $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$. However, by definition 3.37, $\hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \mathbf{1}$ and $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$ entail that $\hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \Longrightarrow \mathtt{B})\urcorner) = \mathbf{1}$. As this contradicts our assumption that $\hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \longrightarrow \mathtt{B})\urcorner) = \mathbf{1}$, we must conclude that there is no cpi $\mathfrak{I}$ such that $\hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \longrightarrow \mathtt{B})\urcorner) = \hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \mathbf{1}$, but $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$.

Moreover, since $\mathbf{1} \in \mathbf{Des}^{+}_{\mathsf{S0}}$ and $\mathbf{0} \in \mathbf{Des}^{-}_{\mathsf{S0}}$, this entails, by definitions 3.39 and 3.41, that $(\mathtt{A} \longrightarrow \mathtt{B}), \mathtt{A} \vDash_{\mathsf{S0}} \mathtt{B}$. $\square$

An alternative proof may be made via a schematic truth table.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{S0}}$ and consider the schematic truth table:

A	B	(A → B)	A	B
0	0	1	0	0
0	1	1	0	1
1	0	0	1	0
1	1	1	1	1

Note that the rows in this table correspond to all possible cpi in relation to A and B. However, there is no cpi $\langle\{\mathbf{0}, \mathbf{1}\}, \iota_{\mathfrak{I}}\rangle$ such that where $\hat{\iota}_{\mathfrak{I}}(\ulcorner(\mathtt{A} \rightarrow \mathtt{B})\urcorner) = \hat{\iota}_{\mathfrak{I}}(\mathtt{A}) = \mathbf{1}$, but $\hat{\iota}_{\mathfrak{I}}(\mathtt{B}) = \mathbf{0}$. Hence, as $\mathbf{1} \in \mathbf{Des}^{+}_{\mathsf{S0}}$ and $\mathbf{0} \in \mathbf{Des}^{-}_{\mathsf{S0}}$, this entails, by definitions 3.39 and 3.41, that $(\mathtt{A} \rightarrow \mathtt{B}), \mathtt{A} \vDash_{\mathsf{S0}} \mathtt{B}$. □

Theorem 3.43. $\neg(\mathtt{A} \vee \mathtt{B}) \vDash_{\mathsf{S0}} (\neg\mathtt{A} \wedge \neg\mathtt{B}) \ \mathbin{\wedge\!\!\!\wedge}\ (\neg\mathtt{A} \wedge \neg\mathtt{B}) \vDash_{\mathsf{S0}} \neg(\mathtt{A} \vee \mathtt{B})$ – i.e., $\neg(\mathtt{A} \vee \mathtt{B}) /\!\!/ (\neg\mathtt{A} \wedge \neg\mathtt{B}) \in \vDash_{\mathsf{S0}}$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{S0}}$ and consider the schematic truth table:

A	B	¬	(A ∨ B)	(¬A	∧	¬B)
0	0	1	0	1	1	1
0	1	0	1	1	0	0
1	0	0	1	0	0	1
1	1	0	1	0	0	0

Note that the rows in this table correspond to all possible cpi in relation to A and B. Hence, we see that all cpi $\langle\{\mathbf{0}, \mathbf{1}\}, \iota_{\mathfrak{I}}\rangle$ are such that $\hat{\iota}_{\mathfrak{I}}(\ulcorner\neg(\mathtt{A} \vee \mathtt{B})\urcorner) = \hat{\iota}_{\mathfrak{I}}(\ulcorner(\neg\mathtt{A} \wedge \neg\mathtt{B})\urcorner)$. Consequently, since $\mathbf{1} \in \mathbf{Des}^{+}_{\mathsf{S0}}$ and $\mathbf{0} \in \mathbf{Des}^{-}_{\mathsf{S0}}$, this entails, by definitions 3.39 and 3.41, that $\neg(\mathtt{A} \vee \mathtt{B}) \vDash_{\mathsf{S0}} (\neg\mathtt{A} \wedge \neg\mathtt{B})$ and $(\neg\mathtt{A} \wedge \neg\mathtt{B}) \vDash_{\mathsf{S0}} \neg(\mathtt{A} \vee \mathtt{B})$. □

Task 3.13. Show that the following inferences are not in $\vDash_{\mathsf{S0}}$:

1. $/ \neg(\neg p \rightarrow p)$
2. $/ \neg(p \rightarrow \neg p)$
3. $(p \rightarrow q) / \neg(p \rightarrow \neg q)$
4. $(p \rightarrow \neg q) / \neg(p \rightarrow q)$

We show just one case.

Theorem 3.44. $(p \to q) \nvDash_{\mathsf{S0}} \neg(p \to \neg q)$. – i.e., $(p \to q)$ / $\neg(p \to \neg q) \notin \vDash_{\mathsf{S0}}$.

Proof. Let the cpi $\langle\{\mathbf{0}, \mathbf{1}\}, \iota_{\mathfrak{I}}\rangle$ be such that $\iota_{\mathfrak{I}}(\text{`}p\text{'}) = \mathbf{0}$. By definition 3.36, this entails $\iota_{\mathfrak{I}}(\text{`}(p \to q)\text{'}) = \mathbf{1}$ and $\iota_{\mathfrak{I}}(\text{`}(p \to \neg q)\text{'}) = \mathbf{1}$. The latter implies, also by definition 3.36, that $\iota_{\mathfrak{I}}(\text{`}(p \to \neg q)\text{'}) = \mathbf{0}$. Now, since $\iota_{\mathfrak{I}}(\text{`}(p \to q)\text{'}) = \mathbf{1}$ and $\iota_{\mathfrak{I}}(\text{`}(p \to \neg q)\text{'}) = \mathbf{0}$, we can infer by definition 3.39, that $(p \to q) \nvDash_{\mathsf{S0}} \neg(p \to \neg q)$. □

Semantics for S1. See chapter 10 of Hannes Leitgeb's script to the course *Logik 1*.

3.8 Derivation System

While an interpretation system provides a semantic account of logical consequence, a *derivation system* or *calculus* provides a *syntactic* account. Instead of assigning meanings to formulae, we specify how formulae may be obtained from others by means of *derivation rules*.

Notion 3.45 (Derivation rule)**.** A **derivation rule** is a rule enabling an expression $\mathtt{E}$ to be written provided other expressions $\mathtt{E}_1, \ldots, \mathtt{E}_m$ are written somewhere else.

A set of derivation rules is *deductive base*.

Definition 3.46 (Deductive base)**.** **DBas**$_{\mathsf{S}}$ is a **deductive base** $\overset{\text{DF}}{=}$ **DBas**$_{\mathsf{S}} \subseteq \{\mathsf{R} : \mathsf{R} \text{ is a derivation rule}\}$.

Deductive bases allow the construction of derivations. A **derivation** may be understood as series of steps some of which are assumptions, and others of which can be obtained from previous steps via a derivation rule.

Notion 3.47 (Derivation)**.** A derivation from a deductive base $\mathbf{DBas}_{\mathrm{S}}$ is a series of expressions:

$$\mathtt{E}_1, \ldots, \mathtt{E}_n.$$

Such that each $\mathtt{E}_i$ is either an assumption or may be obtained from other expressions $\mathtt{E}_{j_1}, \ldots, \mathtt{E}_{j_m}$, where $j_1, \ldots, j_m < n$.

We usually write a derivation as an enumerated list of formulae, and the right of each of which we write within parentheses whether it is an assumption, or whether it follows from other formulae, and its justification.

We will use a special kind of notation for derivations in this style allowing us not only to represent derivations, but also inference rules for a derivation system written in that notation.

Notation 3.48 (Fitch notation)**.** The notation

$$\begin{array}{r|ll}
a_1 & \mathtt{A}_1 & \\
 & \vdots & \\
a_l & \mathtt{A}_l & \\
\cline{2-2}
b_1 & \mathtt{B}_1 & \mathsf{R}_1, b_{1_1}, \ldots, b_{1_{k_1}} \\
 & \vdots & \\
b_n & \mathtt{B}_n & \mathsf{R}_n, b_{n_1}, \ldots, b_{n_{k_n}}
\end{array}$$

describes a derivation on $\mathbf{Fo}_{\mathrm{S}}$ from a deductive base $\mathbf{DBas}_{\mathrm{S}}$ iff:

- $\mathtt{A}_1, \ldots, \mathtt{A}_m, \mathtt{B}_n \in \mathbf{Fo}_{\mathrm{S}0}$,
- $\mathtt{A}_1, \ldots, \mathtt{A}_m$ are the premises and $\mathtt{B}_n$ is the conclusion, and
- each $\mathtt{B}_i$ is obtained from steps $b_{i_1}, \ldots, b_{i_{k_i}}$ by the rule R_i, where $b_{i_1}, \ldots, b_{i_{k_i}} < b_i$.

This notation allows us to outline the notion of a Fitch derivation.

Notion 3.49 (Fitch derivation)**.** A derivation is a Fitch derivation on $\mathbf{Fo}_\mathrm{S}$ from a deductive base $\mathbf{DBas}_\mathrm{S}$ iff it is a derivation of the form specified in notation 3.48 or it is of the form:

$$
\begin{array}{r|ll}
a_1 & \mathtt{A}_1 & \\
 & \vdots & \\
a_l & \mathtt{A}_l & \\
\cline{2-2}
b_1 & \mathtt{B}_1 & \mathsf{R}_1, b_{1_1}, \dots, b_{1_{k_1}} \\
 & \vdots & \\
b_m & \mathtt{B}_m & \mathsf{R}_m, b_{m_1}, \dots, b_{m_{k_m}} \\
 & \mathtt{D} & \\
c_1 & \mathtt{C}_1 & \mathsf{S}_1, c_{1_1}, \dots, c_{1_{k_1}} \\
 & \vdots & \\
c_n & \mathtt{C}_n & \mathsf{S}_n, c_{n_1}, \dots, c_{n_{k_n}}
\end{array}
$$

where:

- $\mathtt{D}$ is a Fitch derivation,
- $\mathtt{A}_1, \dots, \mathtt{A}_l, \mathtt{C}_n \in \mathbf{Fo}_{\mathrm{S}0}$,
- $\mathtt{A}_1, \dots, \mathtt{A}_m$ are the premises and $\mathtt{C}_n$ is the conclusion,
- Each $\mathtt{B}_i$ is obtained from steps $b_{i_1}, \dots, b_{i_{k_i}}$ by the rule R_i, where $b_{i_1}, \dots, b_{i_{k_i}} < b_i$,
- Each $\mathtt{C}_j$ is obtained from steps $c_{j_1}, \dots, c_{j_{k_j}}$ by the rule S_j, where $c_{j_1}, \dots, c_{j_{k_j}} < c_j$.

Thus, Fitch derivations allow for embedding a derivation within another (see next page).

Thus, a *derivation system* or (*logical*) *calculus* $\mathbf{Cal}_{\mathsf{S}}$ on a logic system S consists of a deductive base $\mathbf{DBas}_{\mathsf{S}}$, a set of derivations $\mathbf{Der}_{\mathsf{S}}$, a function $\mathbf{Prem}_{\mathsf{S}}$ establishing the set of premises of $\mathbf{Der}_{\mathsf{S}}$, a function $\mathbf{Conc}_{\mathsf{S}}$ establishing the set of conclusions of $\mathbf{Der}_{\mathsf{S}}$, and a syntactic relation of logical consequence $\vdash_{\mathsf{S}}$.

Definition 3.50 (Derivation system)**.**

$\mathbf{Cal}_{\mathsf{S}}$ is a calculus on S

$$\overset{\text{DF}}{=} \mathbf{Cal}_{\mathsf{S}} = \langle \mathbf{DBas}_{\mathsf{S}}, \mathbf{Der}_{\mathsf{S}}, \mathbf{Prem}_{\mathsf{S}}, \mathbf{Conc}_{\mathsf{S}}, \vdash_{\mathsf{S}} \rangle$$

$: \mathbf{DBas}_{\mathsf{S}}$ is a deductive base on $\mathbf{Fo}_{\mathsf{S}}$ $\wedge\!\!\!\wedge$

$\mathbf{Der}_{\mathsf{S}} = \{\mathbf{D} : \mathbf{D}$ is a derivation on $\mathbf{Fo}_{\mathsf{S}}$ from $\mathbf{DBas}_{\mathsf{S}}\}$ $\wedge\!\!\!\wedge$

$\mathbf{Prem}_{\mathsf{S}} : \mathbf{Der}_{\mathsf{S}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{S}}$ $\wedge\!\!\!\wedge$

$\mathbf{Conc}_{\mathsf{S}} : \mathbf{Der}_{\mathsf{S}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{S}}$ $\wedge\!\!\!\wedge$

$\vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{S}} (\mathbf{Prem}_{\mathsf{S}}(\mathbf{X}) = \mathbf{A} \wedge\!\!\!\wedge \mathbf{Conc}_{\mathsf{S}}(\mathbf{X}) = \mathbf{B})\}.$

Calculus for $\mathsf{S0}$. The deductive base of $\mathsf{S0}$ may be presented as a set of rules governing the use of logical connectives. Each connective is associated with characteristic ways in which statements containing it may be established or used within an argument.

The rules of the system are traditionally divided into introduction and elimination rules. Introduction rules specify the conditions under which a formula containing a given connective may be inferred. Elimination rules, by contrast, describe how information may be extracted from such formulas once they have been established.

The best way to express those rules is in the style of notation 3.48.

Reiteration.

$$\begin{array}{r|ll}
a & \mathtt{A} & \\
& \vdots & \\
& \ldots \mid \mathtt{A} & \text{Reit}, a
\end{array}$$

Negation.

$$\begin{array}{r|ll}
a & \mid \underline{\neg \mathtt{A}} & \\
& \mid \vdots & \\
& \mid \mathtt{B} & \\
& \mid \vdots & \\
b & \mid \neg \mathtt{B} & \\
& \vdots & \\
& \mathtt{A} & \neg^{-}, a\text{–}b
\end{array}
\qquad
\begin{array}{r|ll}
a & \mid \underline{\mathtt{A}} & \\
& \mid \vdots & \\
& \mid \mathtt{B} & \\
& \mid \vdots & \\
b & \mid \neg \mathtt{B} & \\
& \vdots & \\
& \neg \mathtt{A} & \neg^{+}, a\text{–}b
\end{array}$$

Conjunction.

$$\begin{array}{r|ll}
a & (\mathtt{A} \wedge \mathtt{B}) & \\
& \vdots & \\
& \mathtt{A} & \wedge_1^{-}, a
\end{array}
\qquad
\begin{array}{r|ll}
a & (\mathtt{A} \wedge \mathtt{B}) & \\
& \vdots & \\
& \mathtt{B} & \wedge_2^{-}, a
\end{array}
\qquad
\begin{array}{r|ll}
a & \mathtt{A} & \\
& \vdots & \\
b & \mathtt{B} & \\
& \vdots & \\
& (\mathtt{A} \wedge \mathtt{B}) & \wedge^{+}, a, b
\end{array}$$

Disjunction.

$$\begin{array}{r|ll}
a & (\mathtt{A} \vee \mathtt{B}) & \\
& \vdots & \\
b & \mid \underline{\mathtt{A}} & \\
& \mid \vdots & \\
c & \mid \mathtt{C} & \\
& \vdots & \\
d & \mid \underline{\mathtt{B}} & \\
& \mid \vdots & \\
e & \mid \mathtt{C} & \\
& \vdots & \\
& \mathtt{C} & \vee^{-}, a, b\text{–}c, d\text{–}e
\end{array}
\qquad
\begin{array}{r|ll}
a & \mathtt{A} & \\
& \vdots & \\
& (\mathtt{A} \vee \mathtt{B}) & \vee_1^{+}, a
\end{array}
\qquad
\begin{array}{r|ll}
a & \mathtt{B} & \\
& \vdots & \\
& (\mathtt{A} \vee \mathtt{B}) & \vee_2^{+}, a
\end{array}$$

Conditional.

$$\begin{array}{r|ll}
a & (\mathtt{A} \to \mathtt{B}) & \\
 & \vdots & \\
b & \mathtt{A} & \\
 & \vdots & \\
 & \mathtt{B} & \to^-, a, b
\end{array}
\qquad\qquad
\begin{array}{r|l|ll}
a & & \underline{\mathtt{A}} & \\
 & & \vdots & \\
b & & \mathtt{B} & \\
 & \vdots & & \\
 & (\mathtt{A} \to \mathtt{B}) & & \to^+, a\text{–}b
\end{array}$$

Biconditional.

$$\begin{array}{r|ll}
a & (\mathtt{A} \leftrightarrow \mathtt{B}) & \\
 & \vdots & \\
b & \mathtt{A} & \\
 & \vdots & \\
 & \mathtt{B} & \leftrightarrow^-_1, a, b
\end{array}$$

$$\begin{array}{r|ll}
a & (\mathtt{A} \leftrightarrow \mathtt{B}) & \\
 & \vdots & \\
b & \mathtt{B} & \\
 & \vdots & \\
 & \mathtt{A} & \leftrightarrow^-_2, a, b
\end{array}$$

$$\begin{array}{r|l|ll}
a & & \underline{\mathtt{A}} & \\
 & & \vdots & \\
b & & \mathtt{B} & \\
 & \vdots & & \\
c & & \underline{\mathtt{B}} & \\
 & & \vdots & \\
d & & \mathtt{A} & \\
 & \vdots & & \\
 & (\mathtt{A} \leftrightarrow \mathtt{B}) & & \leftrightarrow^+, a\text{–}b, c\text{–}d
\end{array}$$

Thus, we might define $\mathbf{DBas}_{\mathsf{S}}$ as follows:

Definition 3.51 (Deductive base for S0).

$\mathbf{DBas}_{\mathsf{S}} \overset{\text{DF}}{=} \{\mathsf{Reit}, \neg^+, \neg^-, \wedge^+, \wedge^-_1, \wedge^-_2, \vee^+_1, \vee^+_2, \vee^-, \to^+, \to^-, \leftrightarrow^+, \leftrightarrow^-\}.$

As for the derivation system $\mathbf{Cal}_{\mathsf{S0}}$, we will assume that the functions $\mathbf{Prem}_{\mathsf{S0}}$ and $\mathbf{Conc}_{\mathsf{S0}}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 3.52 ($\mathbf{Cal}_{\mathsf{S0}}$).

$\mathbf{Cal}_{\mathsf{S0}}$ is a calculus on S0

$$\begin{aligned}
&\overset{\text{DF}}{=} \mathbf{Cal}_{\mathsf{S0}} = \langle \mathbf{DBas}_{\mathsf{S0}}, \mathbf{Der}_{\mathsf{S0}}, \mathbf{Prem}_{\mathsf{S0}}, \mathbf{Conc}_{\mathsf{S0}}, \vdash_{\mathsf{S0}} \rangle \\
&\ : \mathbf{DBas}_{\mathsf{S0}} \text{ is a deductive base on } \mathbf{Fo}_{\mathsf{S0}} \ \mathbin{\&\!\!\&} \\
&\quad \mathbf{Der}_{\mathsf{S0}} = \{\mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_{\mathsf{S0}} \text{ from } \mathbf{DBas}_{\mathsf{S0}}\} \ \mathbin{\&\!\!\&} \\
&\quad \mathbf{Prem}_{\mathsf{S0}} : \mathbf{Der}_{\mathsf{S0}} \to \wp\mathbf{Fo}_{\mathsf{S0}} \ \mathbin{\&\!\!\&} \\
&\quad \mathbf{Conc}_{\mathsf{S0}} : \mathbf{Der}_{\mathsf{S0}} \to \wp\mathbf{Fo}_{\mathsf{S0}} \ \mathbin{\&\!\!\&} \\
&\quad \vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{S0}} \, (\mathbf{Prem}_{\mathsf{S0}}(\mathbf{X}) = \mathbf{A} \mathbin{\&\!\!\&} \mathbf{Conc}_{\mathsf{S0}}(\mathbf{X}) = \mathbf{B})\}.
\end{aligned}$$

Task 3.14. Show that all inferences of the forms listed on p. 88 are members of $\vdash_{S0}$.

Theorem 3.53. $(\mathtt{A} \longrightarrow \mathtt{B}), \neg\mathtt{B} \vdash_{S0} \neg\mathtt{A}$ – i.e., $(\mathtt{A} \longrightarrow \mathtt{B}), \neg\mathtt{B} \;/\; \neg\mathtt{A} \in \vdash_{S0}$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{S0}$ and consider the schematic derivation:

1	$(\mathtt{A} \longrightarrow \mathtt{B})$	
2	$\neg\mathtt{B}$	
3	$\quad\mid\ \mathtt{A}$	
4	$\quad\mid\ \mathtt{B}$	$\longrightarrow^-$, 1, 3
5	$\quad\mid\ \neg\mathtt{B}$	Reit, 2
6	$\neg\mathtt{A}$	$\longrightarrow^+$, 3–5

Call this derivation '**D**'. It is clear, for all $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{S0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{S0}$ from $\mathbf{DBas}_{S0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{S0}$ with $\ulcorner(\mathtt{A} \longrightarrow \mathtt{B})\urcorner, \ulcorner\neg\mathtt{B}\urcorner \in \mathbf{Prem}_{S0}(\mathbf{D})$ and $\ulcorner\neg\mathtt{A}\urcorner \in \mathbf{Conc}_{S0}(\mathbf{D})$. Therefore, by definition 3.19, we have that $(\mathtt{A} \longrightarrow \mathtt{B}), \neg\mathtt{B} \;/\; \neg\mathtt{A} \in \vdash_{S0}$. □

Theorem 3.54 (Importation). $((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{C}) \vdash_{S0} (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C}))$ – i.e., $((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{C}) \;/\; (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \in \vdash_{S0}$.

Proof. Let $\mathtt{A}, \mathtt{B}, \mathtt{C} \in \mathbf{Fo}_{S0}$ and consider the schematic derivation:

1	$((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{C})$	
2	$\quad\mid\ \mathtt{A}$	
3	$\quad\mid\quad\mid\ \mathtt{B}$	
4	$\quad\mid\quad\mid\ (\mathtt{A} \wedge \mathtt{B})$	$\wedge^+$, 2, 3
5	$\quad\mid\quad\mid\ \mathtt{C}$	$\longrightarrow^-$, 1, 4
6	$\quad\mid\ (\mathtt{B} \longrightarrow \mathtt{C})$	$\longrightarrow^+$, 3–5
7	$(\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C}))$	$\longrightarrow^+$, 2, 6

Call this derivation '**D**'. It is clear, for all $\mathtt{A}, \mathtt{B}, \mathtt{C} \in \mathbf{Fo}_{S0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{S0}$ from $\mathbf{DBas}_{S0}$. This means that there is a derivation in $\mathbf{Der}_{S0}$ with $\ulcorner((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{C})\urcorner \in \mathbf{Prem}_{S0}(\mathbf{D})$ and $\ulcorner(\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C}))\urcorner \in \mathbf{Conc}_{S0}(\mathbf{D})$. Therefore, by definition 3.19, we have that $((\mathtt{A} \wedge \mathtt{B}) \longrightarrow \mathtt{C}) \;/\; (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \in \vdash_{S0}$. □

Note that many of these rules cannot be translated into rules in the sense of notion 3.45. However, they can be recaptured by a logical system on S0 with appropriate definitions of $\mathbf{Der}_{\mathsf{S0}}$, $\mathbf{Prem}_{\mathsf{S0}}$, and $\mathbf{Conc}_{\mathsf{S0}}$.

Calculus for S1. See chapter 11 of Hannes Leitgeb's script to the course *Logik 1*.